Spray Gun



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Lacquering/Polishing Tools

What it is used for

A spray gun applies paint, lacquer, polish, or protective coatings evenly using compressed air or a powered spraying system. Correct setup produces a smooth finish while reducing overspray and material waste.

Main parts

- Gun body with handle and trigger
- Paint or coating cup or container
- Nozzle and fluid needle
- Air cap and adjustment controls
- Air hose or power connection, depending on model
- Cleaning brush and compatible cleaning solution

Before you start

1. Read the manufacturer instructions and identify the coating approved for the gun.
2. Inspect the gun body, trigger, cup, nozzle, and hoses for damage or leaks.
3. Confirm the coating viscosity and strain it if the manufacturer recommends filtering.
4. Check ventilation, remove ignition sources, and wear suitable eye, hand, and respiratory protection.
5. Connect the correct air supply or power source and verify secure fittings.
6. Test the spray pattern on scrap material before coating the workpiece.

How to use

1. Prepare the surface by cleaning, drying, and masking areas that must remain uncoated.
2. Mix and thin the coating according to the product instructions.
3. Fill the container without exceeding its marked capacity and close it securely.
4. Adjust air pressure, fluid flow, and spray pattern according to the gun instructions.
5. Hold the gun at a consistent distance and keep it perpendicular to the surface.
6. Move the gun smoothly across the surface with overlapping passes.
7. Release the trigger at the end of each pass when the gun design permits.
8. Apply additional coats only after the specified drying or flash-off time.
9. Switch off the supply, relieve pressure, and empty unused coating safely.
10. Clean the cup, nozzle, air cap, and fluid passages using approved materials.

⚠ Safety

- Use the coating's safety data sheet to select suitable respiratory and skin protection.
- Operate only in adequate ventilation or an approved spray booth.
- Keep flames, sparks, smoking, and hot surfaces away from flammable coatings and solvents.
- Never point the spray gun at people or test it against skin.
- Depressurize the system before loosening fittings or removing the nozzle.
- Disconnect power or air supply before servicing or clearing blockages.

Care & storage

- Clean the gun immediately after use with a compatible cleaner.
- Use the correct tools to clean the nozzle without enlarging its opening.
- Inspect seals, needles, hoses, and fittings regularly for wear.
- Store the clean, dry gun in a protected location with accessories secured.

Common mistakes to avoid

- Using excessive pressure can cause overspray and uneven coating.
- Holding the gun too close can create runs and heavy deposits.
- Using incompatible thinner or cleaner can damage seals or the coating.
- Leaving coating inside the gun can block passages and damage components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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